# What drives the hourly electricity price in NO1?

## A Gaussian GLM with calendar effects, consumption and reservoir level

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STAT201 project

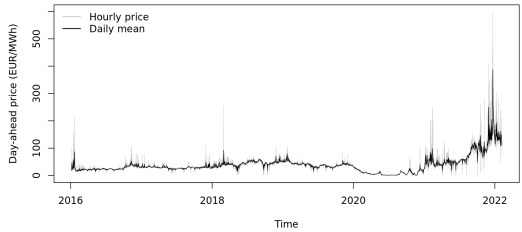
October 2026

# Research question

- Day-ahead price in NO1 (Oslo region), one price per hour
- NO1: much consumption, power mostly from hydro

- **Question:** how much of the hourly price is explained by hour, month, weekday, year, consumption and reservoir level? Which matter most?

- Hypothesis: demand matters, reservoir level even more



# Data

- My master thesis data: Nord Pool prices and consumption, NVE reservoir fill
- Hourly, 04 January 2016 to 06 February 2022: $n = 50{,}855$ **hours**
- Price in EUR/MWh, 11 hours $\leq 0$, max 600.16

**Preparation**

- 2,562 forward-filled prices removed (4.9% in total)
- Weekly reservoir value given to every hour of its ISO week
- Factors from local time; reference: hour 00, January, Monday, 2016

# Method: Gaussian GLM

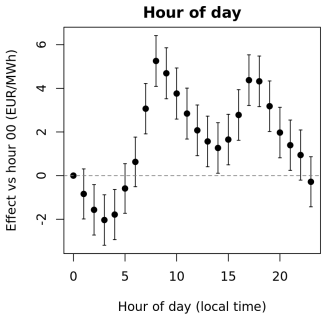
- Price is continuous and can be zero or negative
  $\Rightarrow$ not Poisson, binomial or Gamma; **normal** distribution
- Identity link = multiple linear regression, coefficients in EUR/MWh

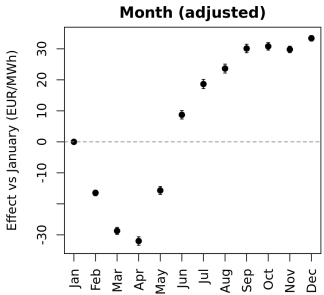
$$y_i = \mathbf{x}_i^T \boldsymbol{\beta} + e_i, \qquad e_i \sim N(0, \sigma^2)$$

```
glm(price ~ hour + month + weekday + year + cons + reservoir,
                        family = gaussian)
```

- Compare: drop-one $F$-test, AIC/BIC, $R^2$; VIF; residuals and ACF
- GAM (`mgcv`) if curves show up

# Key results




- $R^2 = 0.667$, $\hat{\sigma} = 18.75$ EUR/MWh
- Hour 08: $+5.25$ vs midnight
- Sunday: $-4.20$ vs Monday
- 2021: $+48.97$, 2022: $+108.94$ vs 2016
- **Reservoir +10 points:** $-7.1$
- Consumption $+1$ GWh/h: $+2.08$

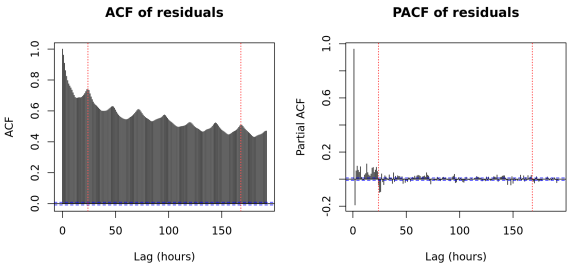
# Which predictors matter most?

Loss when one term is removed from the full model

| Term removed | $\Delta R^2$ (%-points) | $\Delta$AIC |
|---|---|---|
| Year | 44.19 | 42,966 |
| Month | 7.25 | 10,003 |
| Reservoir | 2.86 | 4,192 |
| Hour | 0.27 | 370 |
| Weekday | 0.20 | 295 |
| Consumption | 0.10 | 144 |

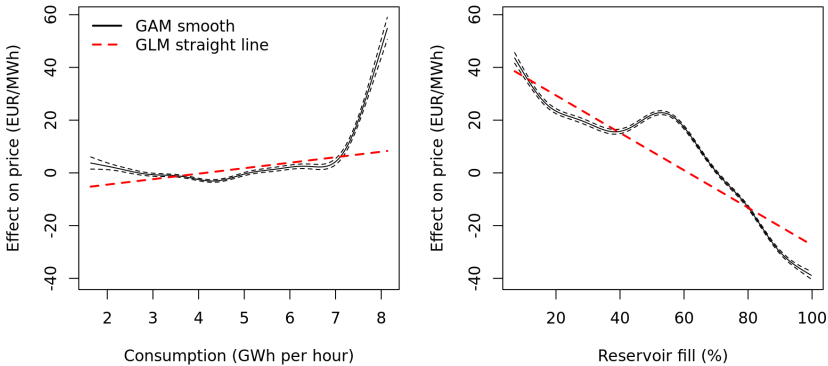
- All $p < 0.001$ with $n = 50{,}855$: look at effect sizes
- Collinearity: month explains 89.4% of reservoir, 76.4% of consumption

# Model fit and diagnostics




- Lag-1 autocorrelation 0.961, Durbin-Watson 0.078
- Variance grows with price, heavy upper tail (2021/22)
- HAC SEs 2.2 to 15.3 times larger
- Reservoir: still clear
- **Consumption:** $p = 0.164$

# GAM: non-linear effects




- Price jumps above 7 GWh/h; reservoir effect strongest at the ends
- AIC $-5{,}222$, $R^2$ 0.667 $\rightarrow$ 0.700; same ranking of predictors

# In plain words, and limitations

**In plain words**

- The year matters most: 2021/22 prices came from outside factors
- Fuller reservoirs, cheaper power: about 7.1 EUR/MWh per 10 points
- Cheapest at night, most expensive around 8 in the morning
- Usage adds little, except in extreme cold hours

**Limitations**

- Strong autocorrelation: usual $p$-values too optimistic
- Non-constant variance, crisis outliers
- Year factor describes, does not explain

# Conclusion

- Gaussian GLM explains $R^2 = 0.667$; year $\gg$ month $>$ reservoir $>$ consumption
- Reservoir effect robust to autocorrelation, consumption effect not; GAM: better fit, same conclusions

# Sources


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[2] John Fox and Sanford Weisberg. *An R Companion to Applied Regression*. Third. Thousand Oaks CA: Sage, 2019.

[3] Nord Pool. *Hourly day-ahead prices and consumption for the bidding zone NO1, 2016–2022*. Raw download used in my master thesis, file `nordpool_2016-01-01_2022-11-30.h5`. Prices in EUR/MWh, volumes in MWh.

[4] NVE (Norges vassdrags- og energidirektorat). *Weekly reservoir filling degree (fyllingsgrad) for NO1*. Column `NO1_Fyllingsgrad_NVE` in my master thesis data set `dataset_timeseries.csv`.

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[6] S. N. Wood. "Fast stable restricted maximum likelihood and marginal likelihood estimation of semiparametric generalized linear models". In: *Journal of the Royal Statistical Society (B)* 73.1 (2011), pp. 3–36.

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[8] Achim Zeileis and Torsten Hothorn. "Diagnostic Checking in Regression Relationships". In: *R News* 2.3 (2002), pp. 7–10.